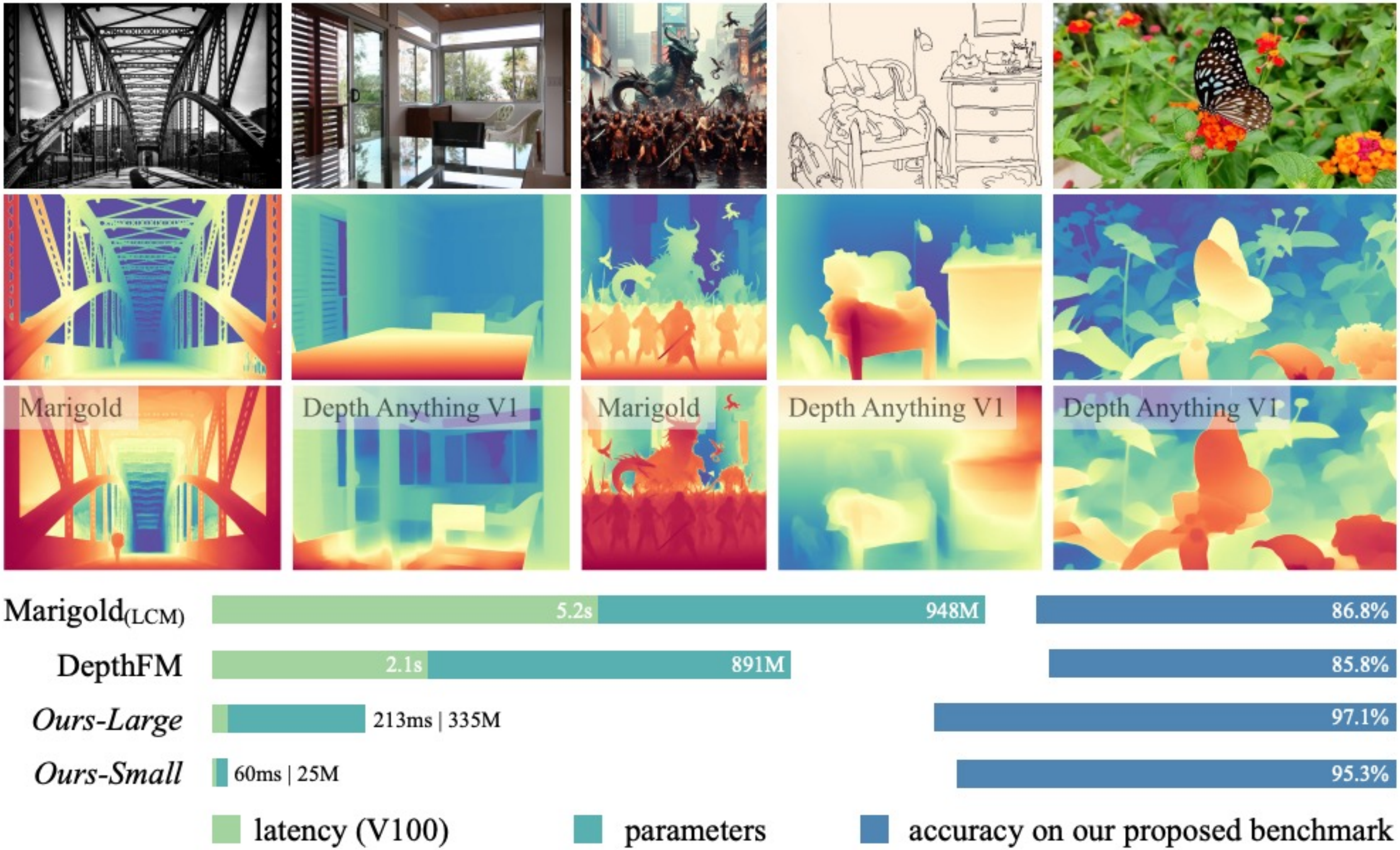
# Depth Anything V2

NeurIPS 2024

이미지처리팀
안종식 허정원 황장훈

# Index

1. Introduction
2. Revisiting the Labeled Data Design of Depth Anything V1
3. Challenges in Using Synthetic Data
4. Key Role of Large-Scale Unlabeled Real Images
5. Overall Framework + Details
6. A New Evaluation Benchmark: DA-2K
7. Experiment
8. Conclusion


Marigold
Depth Anything V1
Marigold
Depth Anything V1
Depth Anything V1
Marigold(LCM)
5.2s
948M
86.8%
DepthFM
2.1s
891M
85.8%
Ours-Large
213ms | 335M
97.1%
Ours-Small
60ms | 25M
95.3%
latency (V100)
parameters
accuracy on our proposed benchmark

# 01

# Introduction

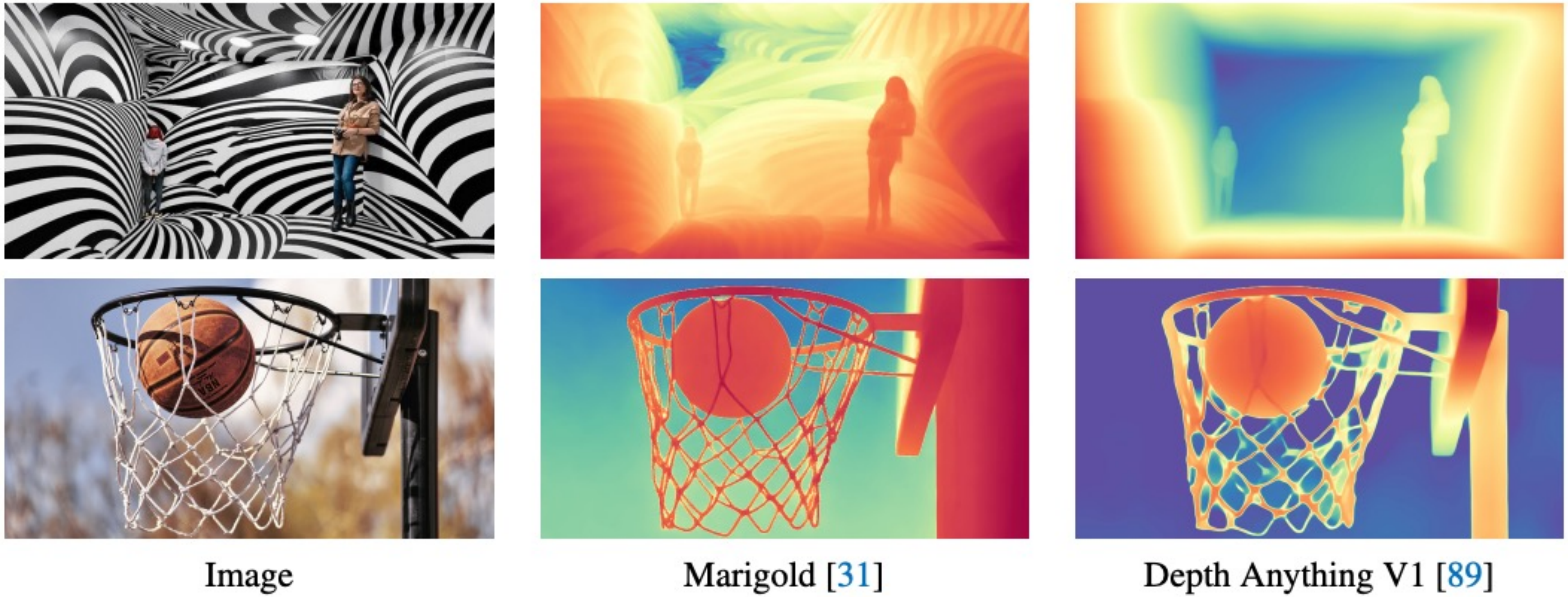



Figure 2: *Robustness* (1st row, the misleading room layout) of Depth Anything V1 and *Fine-grained detail* (2nd row, the thin basketball net) of Marigold.

| Preferable Properties | Fine Detail | Transparent Objects | Reflections | Complex Scenes | Efficiency | Transferability |
|---|---|---|---|---|---|---|
| Marigold [31] | ✓ | ✓ | ✓ | ✗ | ✗ | ✗ |
| Depth Anything V1 [89] | ✗ | ✗ | ✗ | ✓ | ✓ | ✓ |
| Depth Anything V2 (Ours) | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

Table 1: Preferable properties of a powerful monocular depth estimation model.

# 1. Introduction

- **Aim to build a more capable foundation model for MDE**
  - Produce robust predictions for complex scenes, including but not limited to complex layouts, transparent objects (e.g., glass), reflection surfaces (e.g., mirrors, screens), etc.
  - Contain fine details (comparable to the details of Marigold) in the predicted depth maps, including but not limited to thin objects (e.g., chair legs), small holes, etc.
  - Provide varied model scales and inference efficiency to support extensive applications
  - Be generalizable enough to be transferred(i.e., fine-tuned) to downstream tasks

# 1. Introduction

- **Main idea**

No fancy or sophisticated techniques need to be developed.

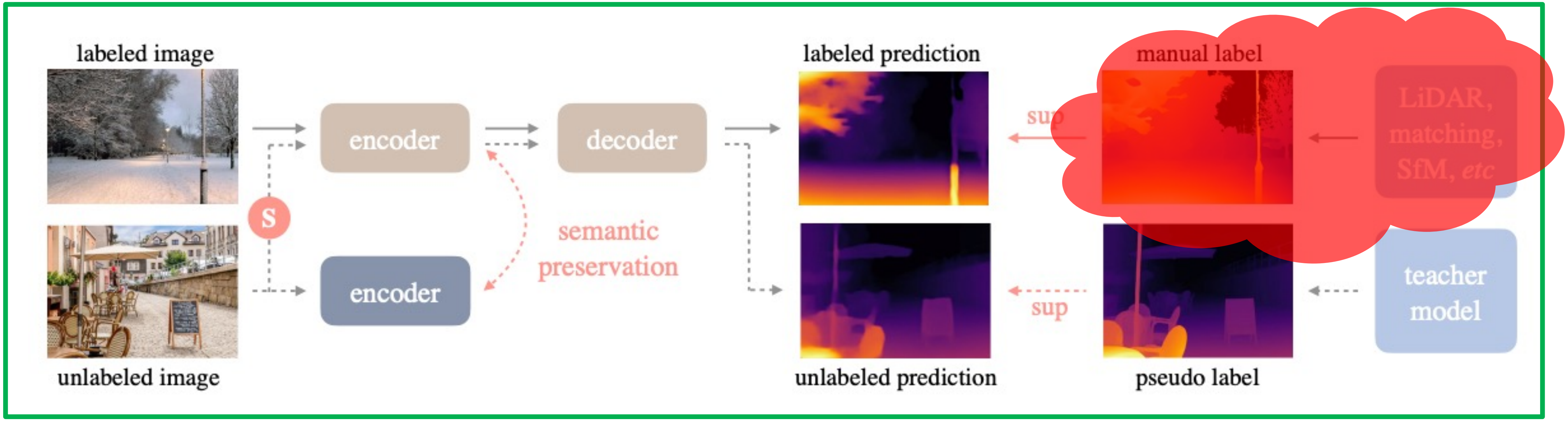
The most critical part is still data.

It is indeed the same as the data-driven motivation of V1, which harnesses large-scale unlabeled data to speed up data scaling-up and increase the data coverage.

In this work, we instead will first **revisit its labeled data design**, and then highlight the key role of unlabeled data.

# 1. Introduction

## Depth Anything V1 pipeline



# 1. Introduction

- **Three key findings**

Q1: Whether the coarse depth of MiDaS or Depth Anything come from the discriminative modeling itself? Is it a must to adopt the heavy diffusion-based modeling manner for fine details?

A1: No, efficient discriminative models can also produce extremely fine details. The most critical modification is replacing all labeled real images with precise synthetic images.

Q2: Why do most prior works still stick to real images, if as A1 mentioned, synthetic images are already clearly superior to real images?

A2: Synthetic images have their drawbacks.

Q3: How to avoid the drawbacks of synthetic images and also amplify its advantages?

A3: Scale up the teacher model that is solely trained on synthetic images, and then teach (smaller) student models via the bridge of large-scale pseudo-labeled real images.

02

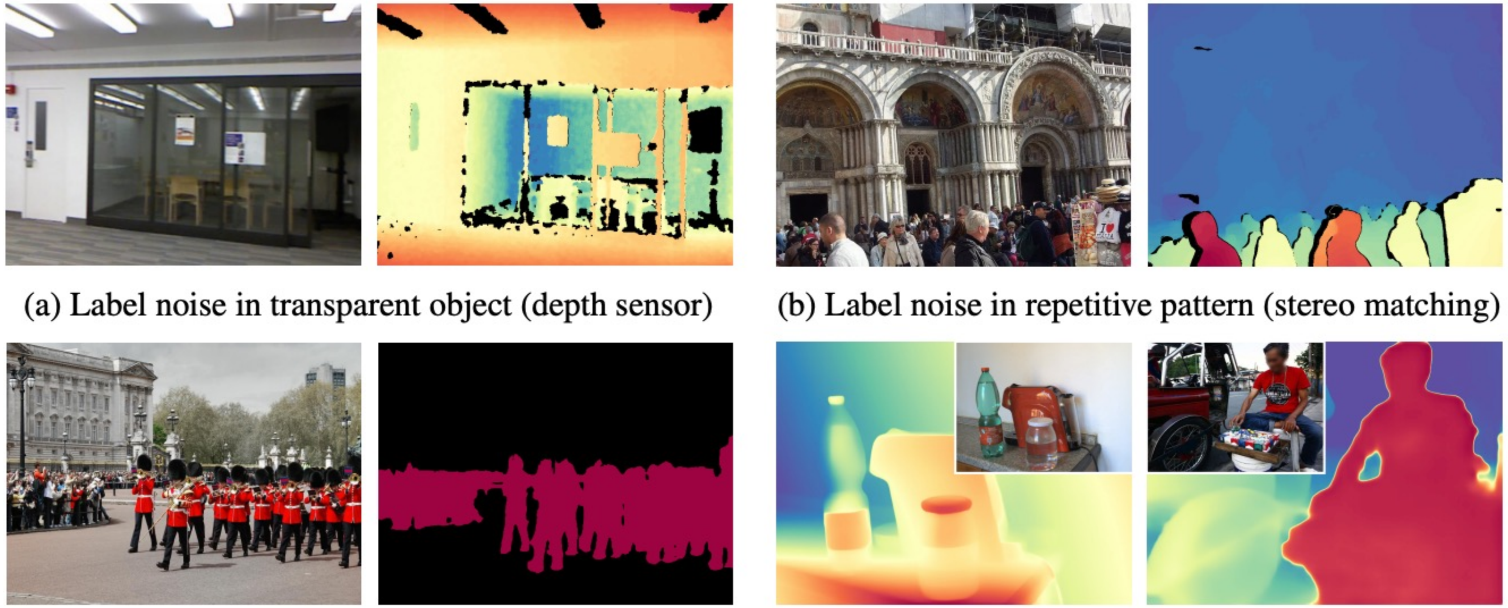
# Revisiting the Labeled Data Design of Depth Anything V1

# 2. Revisiting the Labeled Data Design of Depth Anything V1

Recent studies tend to construct larger-scale training datasets in an effort to enhance estimation performance. However, few studies have critically examined this trend: *is such a huge amount of labeled images truly advantageous?*

**Two disadvantages of real labeled data.**

1. Label noise, i.e., inaccurate labels in depth maps.



(a) Label noise in transparent object (depth sensor)

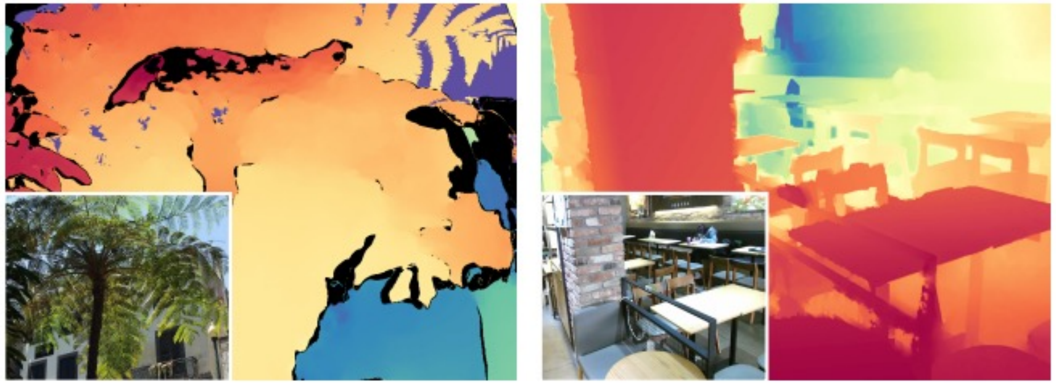
(b) Label noise in repetitive pattern (stereo matching)

(c) Label noise in dynamic objects (SfM)

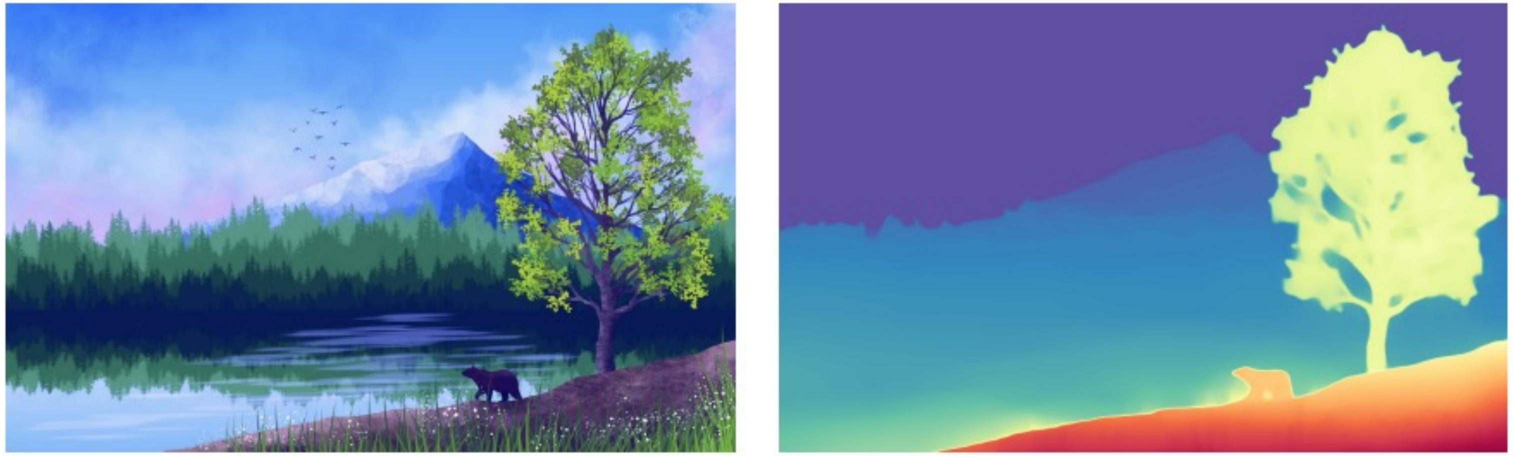
(d) Caused errors in model prediction

# 2. Revisiting the Labeled Data Design of Depth Anything V1

## 2. Ignored details



(a) Coarse depth of real data (HRWSI [83], DIML [14])



(c) Predictions of models trained on labeled real images (middle)

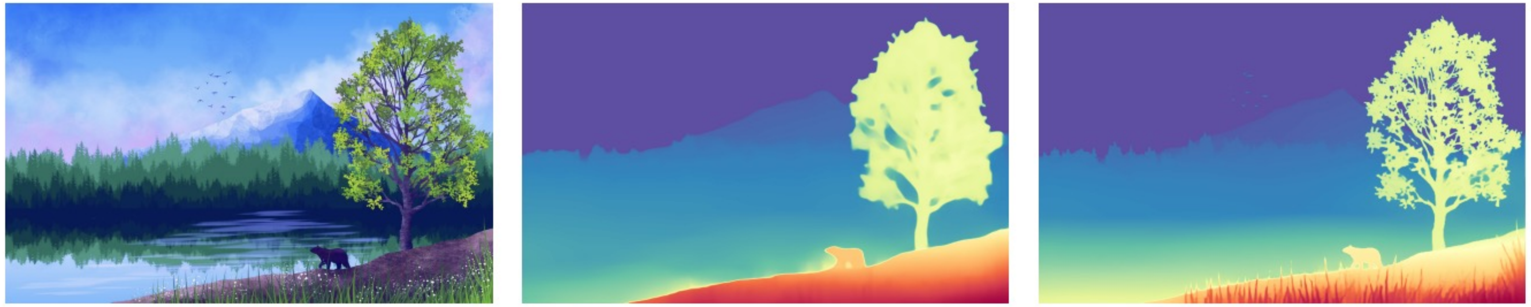
# 2. Revisiting the Labeled Data Design of Depth Anything V1

To overcome the above problems, we extensively check the label quality of synthetic images and note their potential to mitigate the drawbacks discussed above.

**Advantages of synthetic images.**

1) All fine details (e.g., boundaries, thin holes, small objects, etc.) are correctly labeled.
2) We can obtain the actual depth of challenging transparent objects and reflective surfaces.
3) we can quickly enlarge synthetic training images by collecting from graphics engines

(a) Coarse depth of real data (HRWSI [83], DIML [14]) (b) Depth of synthetic data (Hypersim [58], vKITTI [9])



(c) Predictions of models trained on labeled real images (middle) and synthetic images (right)

03

# Challenges in Using Synthetic Data

# 3. Challenges in Using Synthetic Data

If synthetic data are so advantageous, why are real data still dominating MDE?
: Two limitations of synthetic images that hinder them from being easily used in reality

1) Distribution shift between synthetic and real images
- Synthetic images are too "clean" in color and "ordered" in layout, while real images contain more randomness.

(a) Coarse depth of real data (HRWSI [83], DIML [14])

(b) Depth of synthetic data (Hypersim [58], vKITTI [9])

# 3. Challenges in Using Synthetic Data

2) Synthetic images have restricted scene coverage.

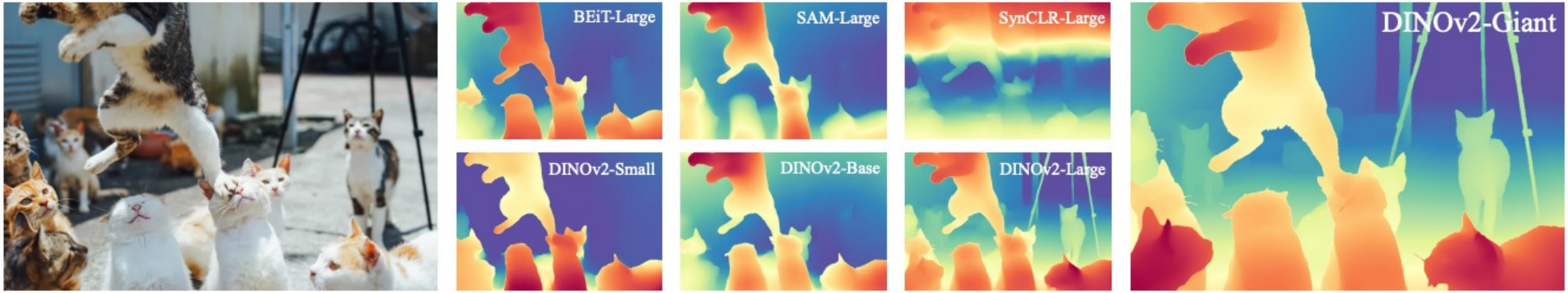
- They are iteratively sampled from graphics engines with pre-defined fixed scene types, e.g., “living room” and “street scene”.

we cannot expect models trained on them to generalize well in real-world scenes like “crowded people”.

In contrast, some real datasets constructed from web stereo images or monocular videos can cover extensive real-world scenes.

# 3. Challenges in Using Synthetic Data

**Therefore, synthetic-to-real transfer is non-trivial in MDE.**
To validate this claim, we conduct a pilot study to learning MDE models solely on synthetic images with four popular pre-trained encoders, including BEiT, SAM, SynCLR, and DINOv2.



# 3. Challenges in Using Synthetic Data

This pilot study seems to give a straightforward solution to employing synthetic data in MDE, i.e., building on the largest DINOv2 encoder, and relying on its **inherent generalization ability.**

However, this naive solution faces two problems.

1)

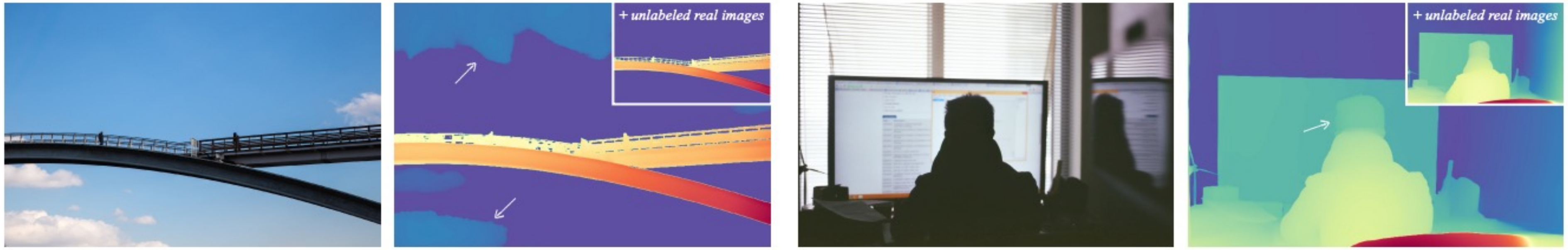


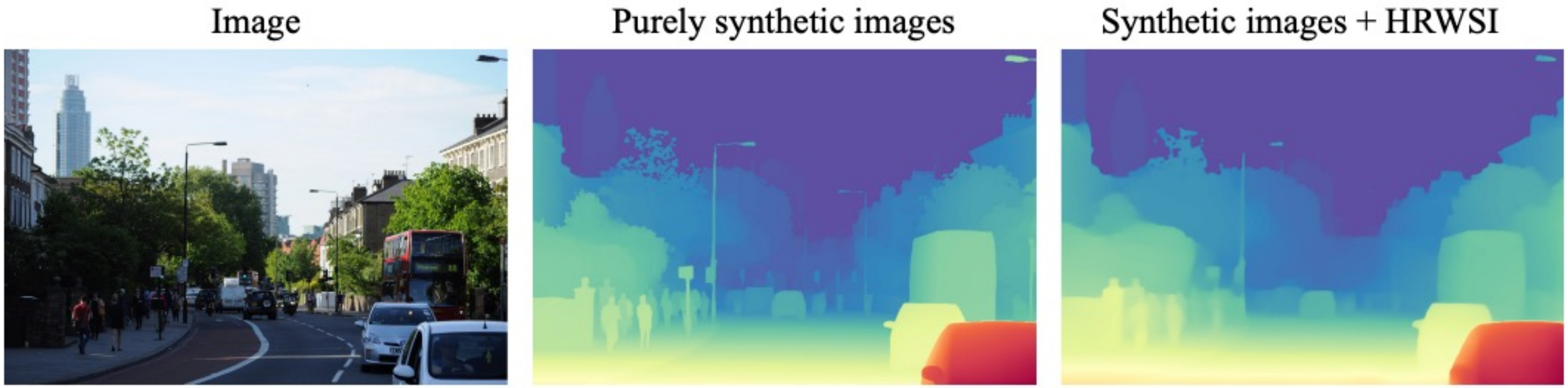

Figure 6: Failure cases of the most capable DINOv2-G model when purely trained on synthetic images. Left: the sky should be ultra far. Right: the depth of the head is not consistent with the body.

2) most applications cannot accommodate the resource-intensive DINOv2-G model (1.3B) in terms of storage and inference efficiency.

# 3. Challenges in Using Synthetic Data

To alleviate the generalization issue,

1) combined training set of real and synthetic images.




2) collect more synthetic images

unsustainable as creating graphic engines mimicking every real-world scenario is intractable.

01

# Key Role of Large-Scale Unlabeled Real Images

# 4. Key Role of Large-Scale Unlabeled Real Images

**Our solution**

1. Our most capable MDE model, based on DINOv2-G, is initially trained purely on high-quality synthetic images.
2. Then it assigns pseudo depth labels on unlabeled real images.
3. Lastly, our new models are solely trained with largescale and precisely pseudo-labeled images.

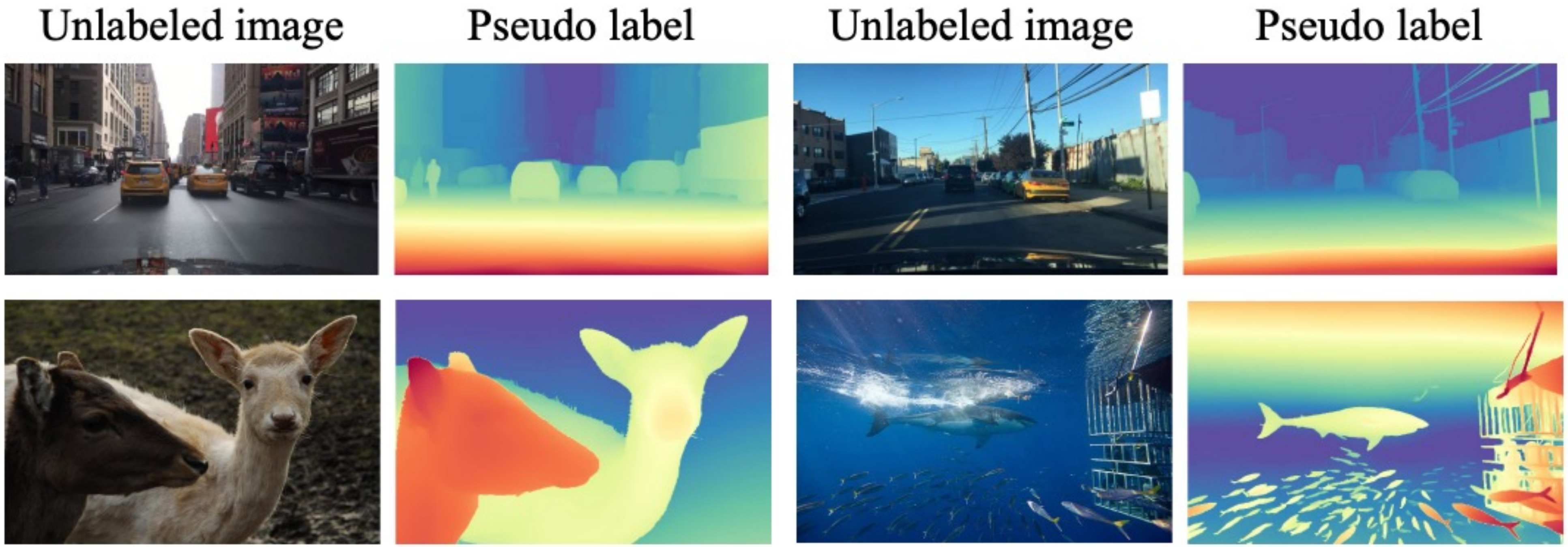
**three perspectives**

- Bridge the domain gap.
- Enhance the scene coverage.
- Transfer knowledge from the most capable model to smaller ones.

# 4. Key Role of Large-Scale Unlabeled Real Images

## Bridge the domain gap

- As aforementioned, due to the distribution shift, directly transferring from synthetic training images to real test images is challenging.
- But if we can leverage extra real images as an intermediate learning target, the process will be more reliable.
- Intuitively, after explicitly training on pseudo-labeled real images, models can be more familiar with real-world data distribution.



# 4. Key Role of Large-Scale Unlabeled Real Images

**Enhance the scene coverage**

- we can easily cover numerous distinct scenes by incorporating large-scale unlabeled images from public datasets.
- By training on sufficient images and scenes, models not only demonstrate stronger zero-shot MDE capability, but they can also serve as better pre-trained sources for downstream related tasks

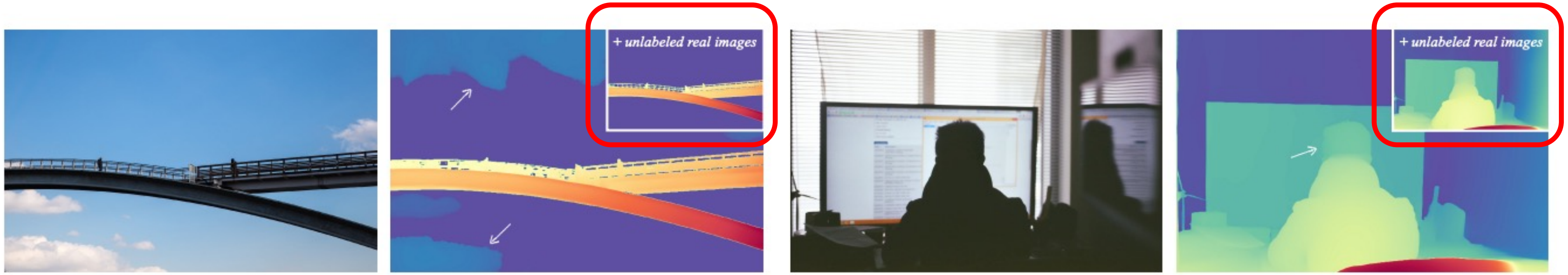
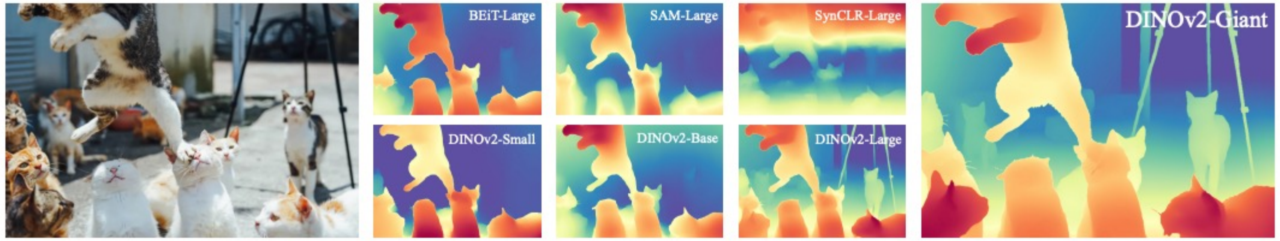



Figure 6: Failure cases of the most capable DINOv2-G model when purely trained on synthetic images. Left: the sky should be ultra far. Right: the depth of the head is not consistent with the body.

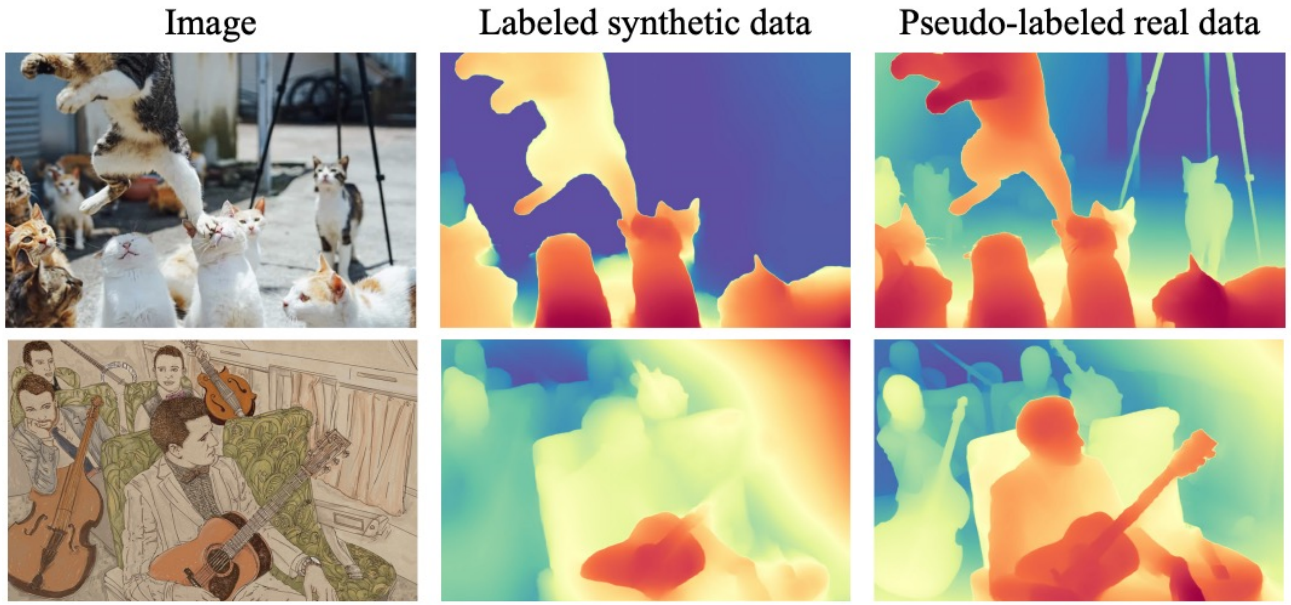
# 4. Key Role of Large-Scale Unlabeled Real Images

## Transfer knowledge from the most capable model to smaller ones

- smaller models cannot directly benefit from synthetic-to-real transfer by themselves.




- However, armed with large-scale unlabeled real images, they can learn to mimic the high-quality predictions of the most capable model, similar to knowledge distillation.

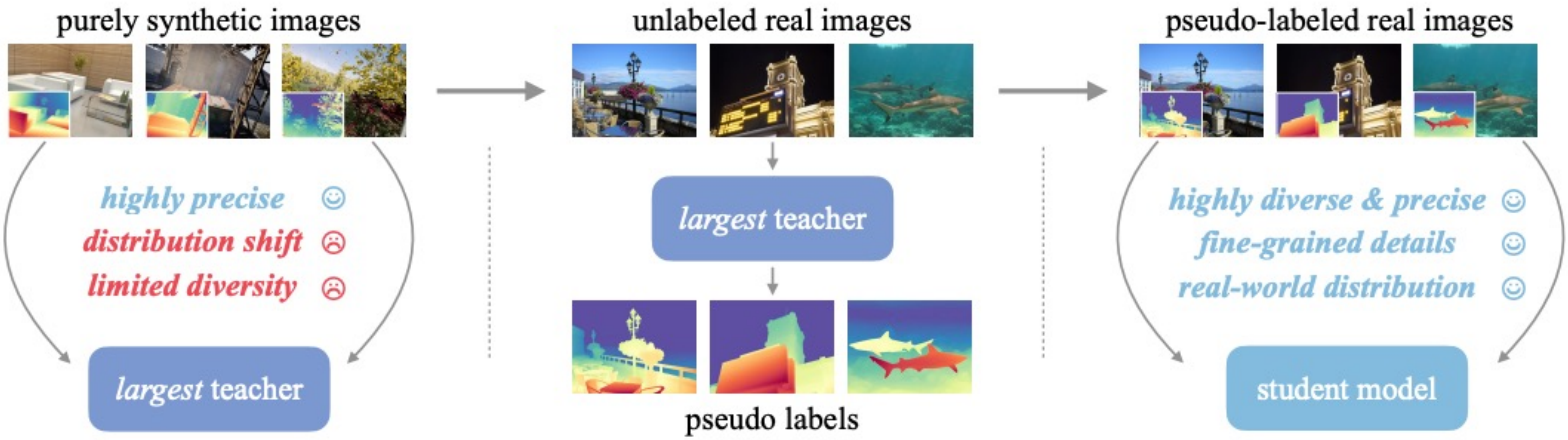



Qualitative comparison of the DINOv2-small-based depth model trained solely on labeled synthetic images and solely pseudo-labeled real images. The robustness is tremendously enhanced.

# 05

# Overall Framework + Details

# 5. Overall Framework + Details



# 5. Overall Framework + Details

Same as V1, for each pseudo-labeled sample, we ignore its top-n-largest-loss regions during training, where n is set as 10%.

We consider them as potentially noisy pseudo labels.

| Dataset | Indoor | Outdoor | # Images |
|---|---|---|---|
| Precise *Synthetic* Images (595K) | | | |
| BlendedMVS [92] | ✓ | ✓ | 115K |
| Hypersim [58] | ✓ | | 60K |
| IRS [77] | ✓ | | 103K |
| TartanAir [79] | ✓ | ✓ | 306K |
| VKITTI 2 [9] | | ✓ | 20K |
| Pseudo-labeled *Real* Images (62M) | | | |
| BDD100K [97] | | ✓ | 8.2M |
| Google Landmarks [81] | | ✓ | 4.1M |
| ImageNet-21K [60] | ✓ | ✓ | 13.1M |
| LSUN [98] | ✓ | | 9.8M |
| Objects365 [65] | ✓ | ✓ | 1.7M |
| Open Images V7 [35] | ✓ | ✓ | 7.8M |
| Places365 [103] | ✓ | ✓ | 6.5M |
| SA-1B [33] | ✓ | ✓ | 11.1M |

Table 7: Our training data sources.

06

# A New Evaluation Benchmark: DA-2K

# 6. A New Evaluation Benchmark: DA-2K

## Limitations in Existing Benchmarks

1. widely adopted test benchmarks are also noisy.

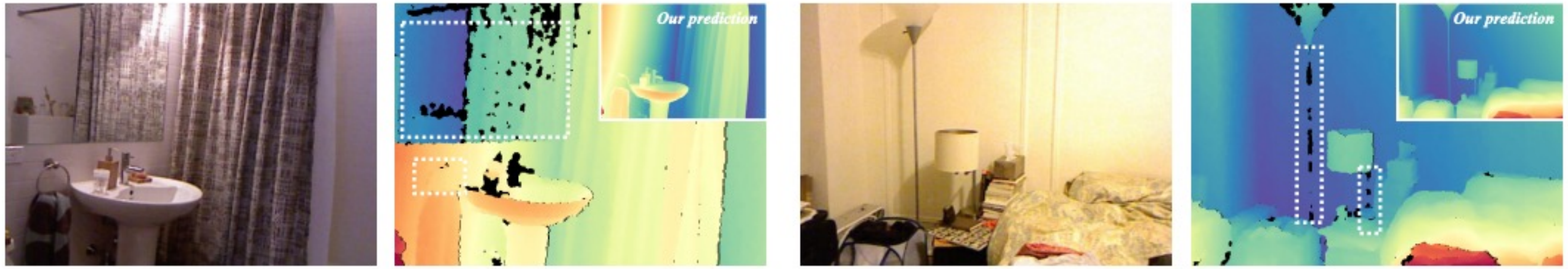



Figure 8: Visualization of widely adopted but indeed noisy test benchmark [70]. As highlighted, the depth of the mirror and thin structures are incorrect (black pixels are ignored). In comparison, our model predictions are accurate. The noise will cause better models instead achieve lower scores.

2. Apart from label noise, another drawback of these benchmarks is limited diversity. Most of them were originally proposed for a single scene.
→ NYU-D focuses on a few indoor rooms, while KITTI only contains several street scenes.

3. The last problem in these existing benchmarks is low resolution.
→ They mostly provide images with a resolution of around 500×500. But with modern cameras, we usually require precise depth estimation for higher-resolution images, e.g., 1000×2000.

# 6. A New Evaluation Benchmark: DA-2K

**DA-2K**

- provide precise depth relationship
- cover extensive scenes
- contain mostly high-resolution images for modern usage

How?

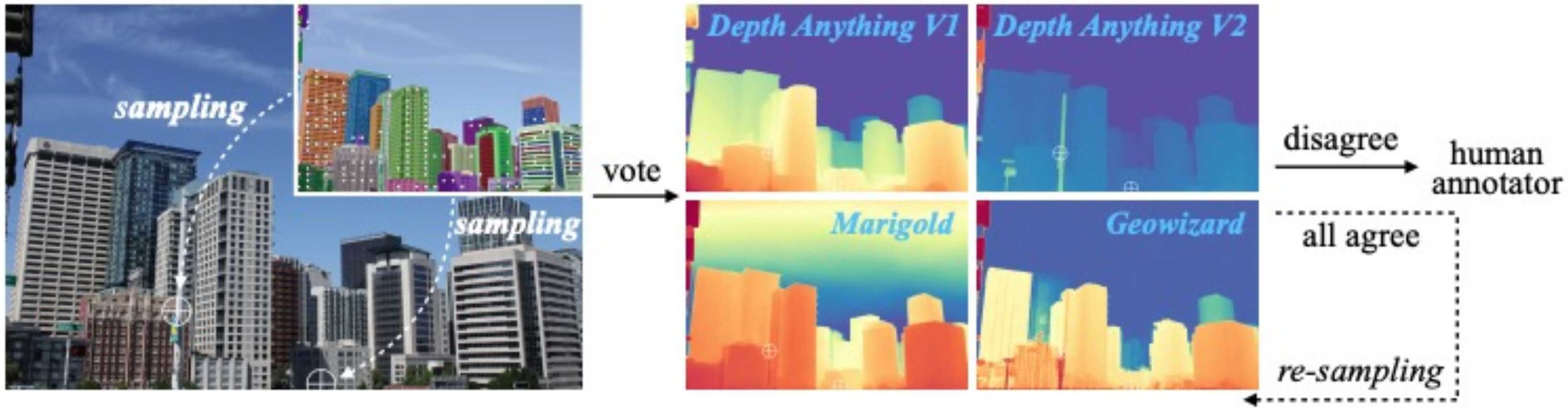
→ we annotate sparse depth pairs for each image.

Generally, given an image, we can select two pixels on it, and decide their relative depth between them (i.e., which pixel is closer).

# 6. A New Evaluation Benchmark: DA-2K

## DA-2K

1. we use SAM to automatically predict object masks
2. Instead of using the masks, we leverage key points (pixels) that prompt out them.
3. We randomly sample two key pixels and query four expert models to vote on their relative depth.
4. If there is disagreement, the pair will be sent to human annotators to decide the true relative depth.




(a) Annotation pipeline

# 6. A New Evaluation Benchmark: DA-2K

## DA-2K

1. To ensure diversity, we first summarize eight important application scenarios of MDE and ask GPT4 to produce diverse keywords related to each scenario.
2. We then use these keywords to download corresponding images from Flickr.
3. Finally, we annotate 1K images with 2K pixel pairs in total.


(b) Encompass 8 scenarios

# 07

# Experiment

# 7. Experiment

## Implementation details

- Follow Depth Anything V1, we use DPT as our depth decoder, built on DINOv2 encoders
- All images are trained at the resolution of 518×518 by resizing the shorter size to 518 followed by a random crop.
- When training the teacher model on synthetic images, we use a batch size of 64 for 160K iterations.
- In the third stage of training on pseudo-labeled real images, the model is trained with a batch size of 192 for 480K iterations.
- We use the Adam optimizer and set the learning rate of the encoder and the decoder as 5e-6 and 5e-5, respectively.

# 7. Experiment

## Zero-Shot Relative Depth Estimation

### Performance on conventional benchmarks

| Method | Encoder | KITTI [24] | | NYU-D [70] | | Sintel [8] | | ETH3D [62] | | DIODE [76] | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | AbsRel | $\delta_1$ | AbsRel | $\delta_1$ | AbsRel | $\delta_1$ | AbsRel | $\delta_1$ | AbsRel | $\delta_1$ |
| MiDaS V3.1 [7] | ViT-L | 0.127 | 0.850 | 0.048 | 0.980 | 0.587 | 0.699 | 0.139 | 0.867 | 0.075 | 0.942 |
| Depth Anything V1 [89] | ViT-S | 0.080 | 0.936 | 0.053 | 0.972 | 0.464 | 0.739 | 0.127 | **0.885** | 0.076 | 0.939 |
| | ViT-B | 0.080 | 0.939 | 0.046 | 0.979 | **0.432** | 0.756 | **0.126** | 0.884 | 0.069 | 0.946 |
| | ViT-L | 0.076 | 0.947 | **0.043** | **0.981** | 0.458 | 0.760 | 0.127 | 0.882 | 0.066 | 0.952 |
| **Depth Anything V2** | ViT-S | 0.078 | 0.936 | 0.053 | 0.973 | 0.500 | 0.718 | 0.142 | 0.851 | 0.073 | 0.942 |
| | ViT-B | 0.078 | 0.939 | 0.049 | 0.976 | 0.495 | 0.734 | 0.137 | 0.858 | 0.068 | 0.950 |
| | ViT-L | **0.074** | 0.946 | 0.045 | 0.979 | 0.487 | 0.752 | 0.131 | 0.865 | 0.066 | 0.952 |
| | ViT-G | 0.075 | **0.948** | 0.044 | 0.979 | 0.506 | **0.772** | 0.132 | 0.862 | **0.065** | **0.954** |

Table 2: Zero-shot *relative* depth estimation. Better: AbsRel $\downarrow$ , $\delta_1$ $\uparrow$. Solely from the metrics, Depth Anything V2 is better than MiDaS, but merely comparable with V1. But indeed, the focus and strengths of our V2 (*e.g.*, fine-grained details, robust to complex layouts, transparent objects, *etc.*) cannot be correctly reflected on these benchmarks. Similar results (*i.e.*, better model but worse score) are also observed in [7, 28].

# 7. Experiment

## Zero-Shot Relative Depth Estimation

Performance on our proposed benchmark DA-2K

| Method | Community Models | | | | Depth Anything V2 (Ours) | | | |
|---|---|---|---|---|---|---|---|---|
| | Marigold [31] | Geowizard [20] | DepthFM [25] | Depth Anything V1 [89] | ViT-S | ViT-B | ViT-L | ViT-G |
| Accuracy (%) | 86.8 | 88.1 | 85.8 | 88.5 | 95.3 | 97.0 | 97.1 | **97.4** |

Table 3: Performance on our proposed DA-2K evaluation benchmark, which encompasses eight representative scenarios. Even our most lightweight model is superior to all other community models.

# 08

# Conclusion

# 8. Conclusion

More powerful foundation model for monocular depth estimation

1) providing robust and fine-grained depth prediction

2) supporting extensive applications with varied model sizes (from 25M to 1.3B parameters)

3) being easily fine-tuned to downstream tasks as a promising model initialization

# Thank you for Listening!